

Is This Enough to Decide? Facilitating Informed Investment Decisions by Visualizing Missing Evidence

Nahyun Lee
Sungkyunkwan University
Suwon, South Korea
leadtheworld@skku.edu

Hyeon Jeon
Seoul National University
Seoul, South Korea
hj@hail.snu.ac.kr

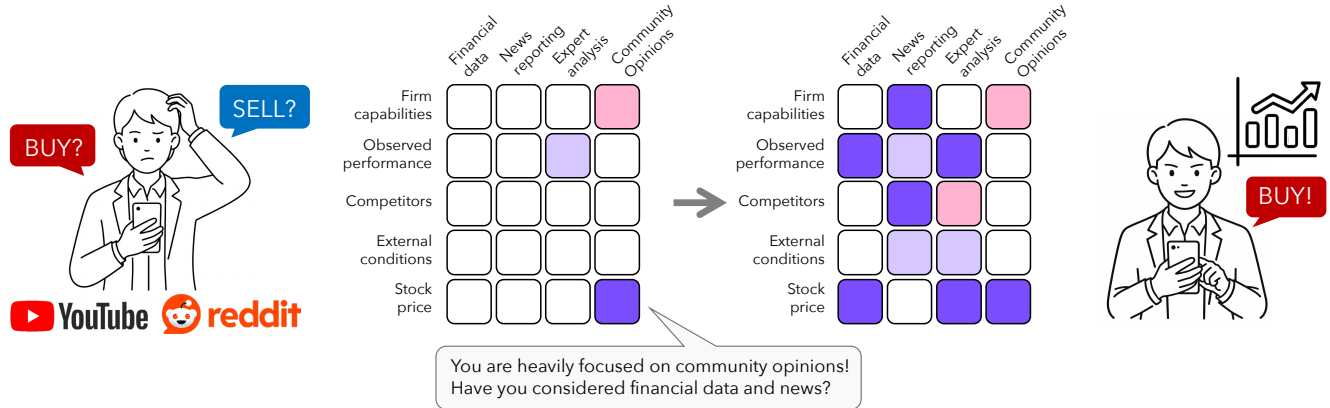


Figure 1: The Evidence Coverage Matrix (ECM) visualizes how collected evidence covers key information, helping users reflect on how informed their decisions are and motivating them to seek missing information.

Abstract

Novice investors rely on online information when making stock investment decisions. However, they are often unaware of whether the evidence they have collected is sufficiently comprehensive to justify their decisions, resulting in investments grounded in incomplete evidence. This can lead to unprofitable investments and hinder financial literacy. To address this challenge, we propose an Evidence Coverage Matrix (ECM) that visualizes how well the collected evidence covers the key information required to evaluate whether a stock is worth investing in. The matrix highlights uncovered areas of the information space, enabling users to reflect on how informed their decisions are and motivating them to seek missing information. Our user study with 10 participants demonstrates that the Coverage Matrix encourages users to collect additional evidence while improving self-reflection and motivation to fill missing information. Based on our findings, we discuss design implications for future systems that support information collection and organization in decision-making.

CCS Concepts

• **Human-centered computing** → **Information visualization.**



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1 Introduction

When making investment decisions, novice investors are often unaware of whether the evidence they have gathered is sufficiently comprehensive to justify their decisions. This difficulty arises in part because investors are exposed to a large volume of information [17]. In such information-rich environments, people tend to selectively consume information that is cognitively inexpensive to process. For example, they often rely on others' interpretations of facts rather than directly inspecting and analyzing the underlying evidence themselves [1, 3, 7, 13]. As a result, novice investors may overlook key indicators that are critical for investment decisions, such as a company's internal capabilities, its competitive landscape, and current stock-price signals. These omissions can lead to investment misjudgments and, over time, hinder the development of novice investors' financial literacy.

To address this challenge, we propose explicitly visualizing which information has been considered and which areas remain overlooked during investment decision-making. To this end, we design the Evidence Coverage Matrix (ECM), a visualization system that

monitors how investors gather evidence and visually summarizes the extent to which their collected information covers key decision perspectives for evaluating a stock. By making it easier for investors to recognize gaps in their evidence collection, the system enables them to reflect on—and iteratively improve—their collection process [3–5, 16, 26]. Furthermore, our system facilitates reflection by providing natural-language feedback in the form of questions that guide users to identify missing yet critical information for decision-making [12, 19, 20, 23, 28].

Through a user study with 10 novice investors, we verify the effectiveness of the ECM in helping users collect more evidence and engage in more effective self-reflection. Participants considered a broader and more balanced range of evidence across diverse perspectives. Moreover, the ECM helped them recognize previously overlooked areas in their decision-making and increased their motivation to explore under-considered evidence in subsequent investment decisions. Based on these findings, we discuss design implications for future decision-support systems that help users collect, organize, and reflect on the sufficiency of information in complex, information-rich decision-making contexts. The contributions of this work are as follows:

- We introduce an **Evidence Coverage Matrix (ECM)**, a visualization system that supports users to be aware of and reflect on the information collected while making investment decisions.
- We conduct a **user study** verifying that the ECM improves novice investors' motivation and capability to collect a more balanced and comprehensive set of evidence.

2 Evidence Coverage Matrix

We introduce the ECM that visualizes how users' investment judgments are supported by evidence across different decision perspectives and sources. We describe the objectives of the matrix, then discuss its design and computational pipeline.

2.1 Objectives

We aim to **(H1)** help novice investors to reflect on the sufficiency and balance of the evidence that grounds their investment decisions. This is because in information-rich environments, novice investors tend to over-rely on information that is cognitively inexpensive to process. As a result, they may struggle to assess whether the evidence they have gathered is sufficiently comprehensive to justify their decisions [1, 13, 17]. Furthermore, we aim to **(H2)** help users reflect on their reasoning and identify remaining evidence gaps to be addressed, thereby facilitating the growth of their financial literacy.

2.2 Visualization Design

We design ECM as a two-dimensional heatmap. Rows represent five core decision perspectives considered when evaluating a stock investment, while columns represent the types of evidence sources used to justify a decision. By doing so, the matrix enables users to visually overview the evidence informing their investment decision across diverse aspects, rather than along a single criterion, and to identify which evidence is emphasized, overlooked, or absent in their reasoning. The detailed design of the matrix is as follows:

Rows and columns. We design the row to represent Damodaran's [6] five decision perspectives—firm capabilities, observed performance, competitors, external conditions, and stock price—that play a central role in company evaluation. This allows users to reflect on whether their judgments are balanced across multiple viewpoints or limited to a subset. We refer to this framework [6] because it is widely accepted in finance [2, 18]. The columns characterize evidence sources by credibility—from factual data to opinions—because the brain processes them differently and information credibility impacts investment returns [11, 15, 29]. We define four tiers: Financial Data, News Reporting, Expert Analysis, and Community Opinions [9, 14, 24]. The list of the perspectives and evidence sources we used in ECM is depicted in Table 1.

Color of each matrix cell. We color each cell to indicate the degree to which the corresponding combination of perspectives and evidence sources is considered during the investment decision-making process. We progressively shift each cell color toward blue as users consume more of the corresponding information by profiling the set of evidence they collect. Lighter shades indicate brief engagement, whereas darker shades indicate more sustained engagement. Empty cells denote perspective–evidence-source pairs for which no evidence has been collected. This color encoding enables users to quickly identify evidence areas that may be overlooked in their reasoning. [10, 25, 27] Sect. 2.3 explains how our system examine and profiles and evidence collected by users.

2.3 Computational Pipeline

We collect users' web browsing logs to profile the set of collected evidence. For each visited page, the system records behavioral signals, including dwell time (time from visit start to end), tap count (number of touches), and idle duration (the longest inactivity period). These signals are used as attention proxies to estimate the likelihood that the page content was meaningfully engaged with. The page is considered to be sufficiently engaged if it satisfies at least one of the following thresholds: dwell time $\geq$ eight seconds, tap count $\geq$ 10, or idle duration $\geq$ five seconds. This threshold is established by examining multiple sources on information retrieval [8]. We then determine the decision-perspective category (row) and evidence-source category (column) to which each piece of content belongs. We prompt an LLM to perform this classification, providing decision guidelines adapted from the original source and using few-shot prompting (Prompts are available in the Appendix 1.1–1.4). A pair of a perspective and an evidence source is considered more thoroughly examined when more user page engagements are assigned to it.

3 User Study

We conduct a controlled user study to examine the effectiveness of ECM in supporting investment decisions of novice investors.

3.1 Objectives and Design

We investigate whether the ECM can **(RQ1)** broaden the range of evidence considered by novice investors (i.e., improve evidence coverage) and **(RQ2)** foster reflective awareness of unexamined information areas and motivate subsequent engagement with those areas. To this end, we ask participants to complete two sessions

Table 1: The list of decision perspectives (top) and evidence sources (bottom) that we use to characterize the rows and columns of ECM and their definition.

Decision Perspective	Description
Firm Capabilities	Internal capabilities possessed by the firm, such as technological expertise, product quality, organizational effectiveness, R&D capacity, and strategy.
Observed Performance	Quantitatively observed performance, such as revenue, profit, growth rates, and reported operational results.
Competitors	Judgments grounded in comparisons with competitors, such as market share, relative strengths and weaknesses, and competitive positioning.
External Conditions	Environmental factors external to the firm’s control, such as political, economic, regulatory, and macro-environmental conditions.
Stock Price	Price-based evaluations and actions, such as current price, volatility, perceptions of over or undervaluation, and stock price trends.
Evidence Source	Description
Financial Data	Structured numerical information, such as financial statements, financial disclosures, and company financial reports.
News Reporting	Fact-focused, event-driven news reporting that primarily conveys factual information about company-related events.
Expert Analysis	Interpretive analysis provided by domain experts, such as analyst reports and professional columns.
Community Opinions	Informal and subjective signals from online investor communities, such as posts, comments, and prevailing sentiment.

making stock investment decisions on different companies. In the first session, participants start by collecting evidence using free web search and making a decision without the support of ECM. They then engage with the matrix and repeat the evidence collection and decision making. In the second session, we repeat the same procedure with a different company. We evaluate the effect of the ECM on the completeness of evidence considered by comparing coverage within each session, and test whether any improvements persist to a subsequent session without further guidance by comparing coverage across sessions (RQ1). We also investigate how the matrix promotes reflective awareness of unexamined information through post-task questionnaires (RQ2).

Procedure. After participants sign the consent form, they watch a five-minute video tutorial explaining the overall system workflow. Participants then complete a questionnaire assessing their investment experience and their perceived importance of different information types for stock investment (Table 1) (Full questionnaire available in the Appendix 2.1–2.3). Participants complete two sessions, each followed by a post-session survey that primarily assesses their reflective awareness (RQ2). After each session, participants completed a post-session questionnaire assessing their evidence awareness, decision-making process, and information use. The questionnaire assesses important facets of reflective awareness [21, 22], including overlooked evidence, perceived importance of different information types, reliance on different evidence sources, and intentions for future information exploration, and includes open-ended responses. Finally, we interview participants, focusing on their experience with the Evidence Coverage Matrix and their awareness of the coverage of the evidence they collected.

Participants. We recruit 10 novice investors (7 males and 3 females, aged 19–37 [25.3 ± 5.7]) who participated in the study. We require all participants to have less than one year of stock investment experience.

Selected companies. We use NVIDIA (NVDA) and Tesla (TSLA) for the companies we ask participants to make investment decisions for. We select these two companies because they are globally well-known technology stocks that have both experienced substantial recent price increases, yielding comparable contexts for cross-session comparison. Moreover, both companies are associated with abundant and diverse online information, making them

suitable targets for examining the effectiveness of ECM in revealing imbalances and gaps in collected evidence. To mitigate order effects, half of the participants evaluate NVIDIA in Session 1 and Tesla in Session 2, while the remaining participants complete the sessions in the reverse order.

Facilitating reflection. We support participants’ reflection on their investment decisions by allowing them to interact with ECM. After participants make an investment decision, the system prompts them to articulate the core evidence underlying that decision. We then map their responses onto the ECM by decision perspective (rows) and evidence source (columns). Based on this mapping, a cell is highlighted in red when the corresponding evidence is mentioned but was not obtained on the web during the collection process, and in purple when it was both collected and explicitly mentioned. If collected but not mentioned, it is highlighted in blue as with the original design of the matrix. This encoding enables participants to see which aspects of their judgment are grounded in collected evidence versus generated from their own prior knowledge or assumptions.

We also prompt the matrix to generate natural language feedback based on its current state. It (1) summarizes the areas that have been considered and those that remain unexamined, and (2) highlights the unexamined decision perspectives or evidence types. Finally, it (3) provides suggestions for further exploration, but does not prescribe specific investment actions such as buying or selling. We explicitly prompt the LLM not to provide investment advice or trading recommendations. This feedback is given between the first investment decision and the second phase of evidence collection of each session [4, 23, 28].

3.2 Quantitative Results

We report our quantitative results that answer RQ1 and RQ2.

RQ1: Evidence coverage. We examine whether the ECM increases the breadth and diversity of evidence considered during investment decision making. Coverage score reflects the comprehensiveness of engagement with diverse perspectives and evidence sources, not decision quality. The main findings are as follows.

Finding 1) The matrix prompts an immediate increase in evidence coverage. We compare how evidence coverage changes within sessions before and after interacting with the ECM. We compute

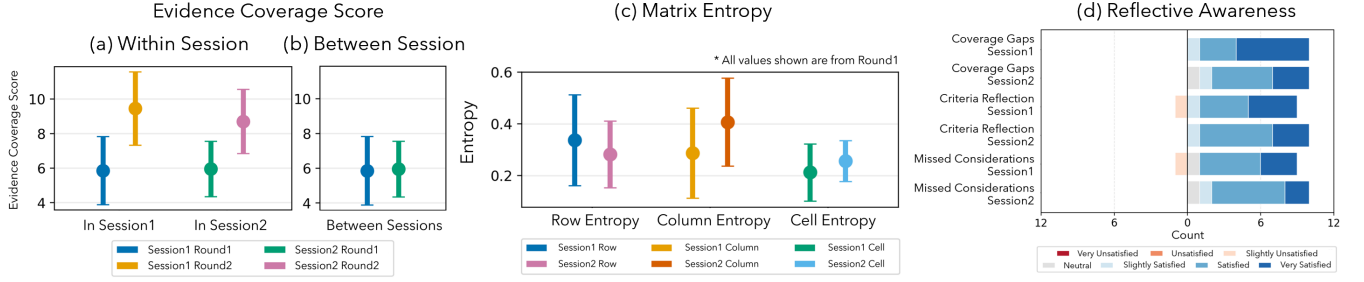


Figure 2: Quantitative results of the user study. (a) Evidence Coverage Scores within sessions. (b) Evidence Coverage Scores between sessions. (c) Row, column, and cell entropy values for Round 1 in each session. (d) Post-session survey results on reflective awareness across sessions.

the coverage score as a weighted sum of cell states, assigning 0.5 to short visits, 1.0 to long visits, and an additional 0.5 when the evidence is mentioned. As a result (Fig. 2a), we find that coverage becomes significantly higher after interacting with the matrix, for both session 1 ($t(9) = 5.78, p < .001, d = 1.83$) and 2 ($t(9) = 4.47, p = .0016, d = 1.41$).

Finding 2) The matrix does not provide transferrable learning effect. We compare changes in evidence coverage between Sessions 1 and 2, both before interacting with the ECM. By doing so, we examine how the experience in Session 1 impacts the coverage of evidence in Session 2. As a result (Fig. 2b), we find no significant difference between the two conditions ($t(9) = 0.12, p = .91, d = 0.04$), indicating that a single prior exposure did not lead to a measurable increase in evidence coverage.

Finding 3) The matrix may increase the diversity of explored evidence. We compare the diversity of evidence explored in Sessions 1 and 2. Given a set of evidence items E , we compute its entropy as $S(E) = -\sum_{e \in E} P_e \log P_e$, where P_e denotes the proportion of item e in E . We partition evidence into sets according to their assignment to (1) rows, (2) columns, and (3) individual cells of the matrix, and compute entropy for each partition to compare diversity across sessions. As a result, we find no significance difference for rows ($t(9) = 0.47, p = .65, d = 0.15$), columns ($t(9) = -1.10, p = .30, d = -0.35$), or cells ($t(9) = -0.74, p = .48, d = -0.23$). However, the medium effect size (d -value) for column and cell entropy suggests that the ECM may cause a statistically significant increase in these entropies.

RQ2: Reflective awareness. We assessed participants' reflective awareness of unexamined information using the post-task questionnaire items. Participants report overall high levels of reflective awareness in both sessions (Session 1: $M = 6.10, SD = 1.04$; Session 2: $M = 6.03, SD = 0.89$), recording mean scores over five (slightly satisfied; Fig. 2d).

3.3 Qualitative Findings

Our qualitative findings resonate with quantitative results, further verifying the effectiveness of the ECM. Several participants reported that they often buy stocks popular in online communities based on intuition; however, after engaging with the matrix's feedback, they explored previously unconsidered information areas and made

investment decisions based on their own reasoning. One participant noted that they had focused on short-term investing and felt anxious about price fluctuations, but that the matrix could support more confident long-term investment by clarifying their reasoning.

3.4 Discussions

We find that the ECM improves evidence coverage in investment decision-making and may broaden the diversity of perspectives considered. Although no behavioral transfer is observed across sessions, consistently high reflective awareness suggests that the matrix supports recognition of unexamined information. These results indicate that reflective visualizations like the ECM facilitate immediate self-reflection but may require repeated exposure to generalize across decisions. Importantly, greater coverage does not necessarily imply better decisions, as additional information may be redundant or low quality; future work should examine its relationship with decision quality. As currently designed, the ECM focuses on short-term reflective support; exploring designs for sustained, long-term engagement remains future work.

4 Conclusion

We address the challenge novice investors face in assessing whether the evidence they collect is sufficiently comprehensive to justify their investment decisions. We propose the ECM, a reflective visualization that makes explicit how collected evidence covers key decision perspectives and evidence sources. Our user study verifies that the matrix contributes in enhancing the coverage of collected evidence and fosters reflective awareness of unexamined information. We highlight the value of reframing decision support from action recommendation to reflective support, and argue that reflective visualizations can help users reason about what has been considered and what remains overlooked in information-rich decision-making contexts.

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References

- [1] Anthony Bellofatto, Catherine D'Hondt, and Rudy De Winne. 2018. Subjective financial literacy and retail investors' behavior. *Journal of Banking & Finance* 92 (2018), 168–181. doi:10.1016/j.jbankfin.2018.05.004
- [2] Mauro Bini. 2021. Business Valuation and Fundamental Analysis. *Business Valuation OIV Journal* 3, 1 (2021), 16–23.
- [3] Sayan Chaudhry and Chinmay Kulkarni. 2021. Design Patterns of Investing Apps and Their Effects on Investing Behaviors. In *Proceedings of the 2021 ACM Designing Interactive Systems Conference (Virtual Event, USA) (DIS '21)*. Association for Computing Machinery, New York, NY, USA, 777–788. doi:10.1145/3461778.3462008
- [4] Rimika Chaudhury and Parmit K. Chilana. 2025. Designing Visual and Interactive Self-Monitoring Interventions to Facilitate Learning: Insights From Informal Learners and Experts. *IEEE Transactions on Visualization and Computer Graphics* 31, 2 (2025), 1542–1556. doi:10.1109/TVCG.2024.3366469
- [5] Patrick Crain, Jaewook Lee, Yu-Chun Yen, Joy Kim, Alyssa Aiello, and Brian Bailey. 2023. Visualizing Topics and Opinions Helps Students Interpret Large Collections of Peer Feedback for Creative Projects. *ACM Trans. Comput.-Hum. Interact.* 30, 3, Article 49 (June 2023), 30 pages. doi:10.1145/3571817
- [6] Aswath Damodaran. 2011. *The Little Book of Valuation: How to Value a Company, Pick a Stock, and Profit*. John Wiley & Sons, Hoboken, NJ, USA.
- [7] Seoyoung Doh, Hyeon Jeon, Sungbok Shin, Ghulam Jilani Quadri, Nam Wook Kim, and Jinwook Seo. 2025. Understanding Bias in Perceiving Dimensionality Reduction Projections. arXiv:2507.20805 [cs.HC]
- [8] Nir Grinberg. 2018. Identifying Modes of User Engagement with Online News and Their Relationship to Information Gain in Text. In *Proceedings of the 2018 World Wide Web Conference (Lyon, France) (WWW '18)*. International World Wide Web Conferences Steering Committee, 1745–1754. doi:10.1145/3178876.3186180
- [9] Timo Harjunemi. 2022. The "Hierarchy of Credibility" among Economic Experts: Journalists' Perceptions of Experts with Varying Institutional Affiliations. *Journalism Practice* 16, 8 (2022), 1635–1652. doi:10.1080/17512786.2021.1910985
- [10] C.G. Healey and J.T. Enns. 1999. Large datasets at a glance: combining textures and colors in scientific visualization. *IEEE Transactions on Visualization and Computer Graphics* 5, 2 (1999), 145–167. doi:10.1109/2945.773807
- [11] Aleksandra Kawiak, Grzegorz M. Wojcik, Piotr Schneider, Lukasz Kwasniewicz, and Adam Wierzbicki. 2020. Whom to Believe? Understanding and Modeling Brain Activity in Source Credibility Evaluation. *Frontiers in Neuroinformatics* 14 (2020), 607853. doi:10.3389/fninf.2020.607853
- [12] Hyung-Kwon Ko, Hyeon Jeon, Gwanmo Park, Dae Hyun Kim, Nam Wook Kim, Juho Kim, and Jinwook Seo. 2024. Natural Language Dataset Generation Framework for Visualizations Powered by Large Language Models. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (Honolulu, HI, USA) (CHI '24)*. Association for Computing Machinery, New York, NY, USA, Article 843, 22 pages. doi:10.1145/3613904.3642943
- [13] Benson Tsz Kin Leung. 2020. Limited cognitive ability and selective information processing. *Games and Economic Behavior* 120 (2020), 345–369. doi:10.1016/j.geb.2020.01.005
- [14] Kanika Marwaha and Sangeeta Arora. 2016. Information Sources Credibility and Its Impact on Individual Investor Behavior: An Empirical Analysis. *Rai Management Journal* 13, 1 (2016), 14–23.
- [15] Daniel J. Mirny and Stephen A. Spiller. 2025. Source Memory Is More Accurate for Opinions than for Facts. *Journal of Consumer Research* 52, 4 (2025), 779–799. doi:10.1093/jcr/ucaf007
- [16] Lace M. Padilla, Sarah H. Creem-Regehr, Mary Hegarty, and Jeanine K. Stefanucci. 2018. Decision Making with Visualizations: A Cognitive Framework across Disciplines. *Cognitive Research: Principles and Implications* 3, 1, Article 29 (2018). doi:10.1186/s41235-018-0120-9
- [17] Meng Peng, Zhe Xu, and Huan Huang. 2021. How Does Information Overload Affect Consumers' Online Decision Process? An Event-Related Potentials Study. *Frontiers in Neuroscience* 15, Article 695852 (2021). doi:10.3389/fnins.2021.695852
- [18] Michael E. Porter. 1980. *Competitive Strategy: Techniques for Analyzing Industries and Competitors*. Free Press, New York, NY.
- [19] Leon Reicherts, Zelun Tony Zhang, Elisabeth von Oswald, Yuanting Liu, Yvonne Rogers, and Mariam Hassib. 2025. AI, Help Me Think—but for Myself: Assisting People in Complex Decision-Making by Providing Different Kinds of Cognitive Support. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, NY, USA, Article 255, 19 pages. doi:10.1145/3706598.3713295
- [20] Wendy Roldan, Ziyue Li, Xin Gao, Sarah Kay Strickler, Allison Marie Hishikawa, Jon E. Froehlich, and Jason Yip. 2021. Pedagogical Strategies for Reflection in Project-based HCI Education with End Users. In *Proceedings of the 2021 ACM Designing Interactive Systems Conference (Virtual Event, USA) (DIS '21)*. Association for Computing Machinery, New York, NY, USA, 1846–1860. doi:10.1145/3461778.3462113
- [21] Gregory Schraw and Rayne Sperling Dennison. 1994. Assessing Metacognitive Awareness. *Contemporary Educational Psychology* 19, 4 (1994), 460–475. doi:10.1006/ceps.1994.1033
- [22] Laura Schulz, Stephen M. Fleming, and Peter Dayan. 2023. Metacognitive computations for information search: Confidence in control. *Psychological Review* 130, 3 (2023), 604–639. doi:10.1037/rev0000401
- [23] Noah Shinn, Federico Cassano, Edward Berman, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. 2023. Reflexion: Language Agents with Verbal Reinforcement Learning. arXiv:2303.11366 [cs.AI]
- [24] Nadine Strauß. 2019. Financial journalism in today's high-frequency news and information era. *Journalism* 20, 2 (2019), 274–291. doi:10.1177/1464884917753556 PMID: 30662364.
- [25] Maoyuan Sun, Yue Ma, Yuanxin Wang, Tianyi Li, Jian Zhao, Yujun Liu, and Ping-Shou Zhong. 2022. Toward Systematic Considerations of Missingness in Visual Analytics. arXiv:2108.04931 [cs.HC]
- [26] Dishita Turakhia, Zane Mroue, Peiling Jiang, and Stefanie Mueller. 2024. Generating Reflection Prompts in Self-Directed Learning Activities with Generative AI. An MIT Exploration of Generative AI. <https://mit-genai.pubpub.org/pub/kju0447a>
- [27] Colin Ware. 2020. *Information Visualization: Perception for Design* (4 ed.). Morgan Kaufmann, San Francisco, CA, USA. doi:10.1016/C2016-0-02395-1
- [28] Kefan Xu, Xinghui (Erica) Yan, Myeonghan Ryu, Mark W. Newman, and Rosa I. Arriaga. 2024. Understanding the Effect of Reflective Iteration on Individuals' Physical Activity Planning. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (Honolulu, HI, USA) (CHI '24)*. Association for Computing Machinery, New York, NY, USA, Article 106, 17 pages. doi:10.1145/3613904.3641937
- [29] Zhen Yu, Michael D. Wang, Xiangdong Wei, and Jie Lou. 2023. News Credibility and Influence within the Financial Markets. *Journal of Behavioral Finance* 24, 2 (2023), 238–257. doi:10.1080/15427560.2021.1974443